

SQL (week-0)

Question 1:

Basic Filtering – High-Value Products

Query:

```
SELECT event_id, store_id, product_code, base_price, promo_type FROM fact_events WHERE  
base_price > 1000;
```

Output:

	event_id	store_id	product_code	base_price	promo_type
▶	a1503f	STCBE-1	P15	3000	500 Cashback
	6b2afc	STCBE-4	P08	1190	BOGOF
	8f25a6	STHYD-6	P15	3000	500 Cashback
	0f422c	STBLR-0	P14	1020	BOGOF
	7fc923	STTRV-0	P08	1190	BOGOF
	ca7298	STCBE-2	P15	3000	500 Cashback
	bf33ae	STMYS-0	P14	1020	BOGOF
	4f0587	STMDU-0	P08	1190	BOGOF
	fa5b45	STVJD-1	P08	1190	BOGOF
	7e9777	STVJD-1	P08	1190	BOGOF
	761785	STBLR-4	P14	1020	BOGOF
	eb3bea	STVSK-2	P15	3000	500 Cashback
	627200	STCBE-4	P14	1020	BOGOF
	c8ce63	STMDU-0	P15	3000	500 Cashback
	57a7bc	STHYD-5	P14	1020	BOGOF
	70e65e	STCBE-0	P15	3000	500 Cashback
	32efe6	STVSK-3	P08	1190	BOGOF
	d834f0	STMYS-3	P08	1190	BOGOF
	268ca2	STVJD-0	P15	3000	500 Cashback
	1548f8	STVSK-4	P15	3000	500 Cashback

Question 2:

Sorting Promotional Events

Query:

```
SELECT event_id, product_code, promo_type, `quantity_sold(before_promo)`,  
`quantity_sold(after_promo)` FROM fact_events WHERE `quantity_sold(after_promo)` > 100 ORDER BY  
`quantity_sold(after_promo)` DESC;
```

Output:

	event_id	product_code	promo_type	quantity_sold(before_promo)	quantity_sold(after_promo)
▶	e93b62	P04	BOGOF	513	2067
	1a1a02	P04	BOGOF	465	2064
	fe94ae	P04	BOGOF	450	1984
	4eb35e	P04	BOGOF	483	1907
	fe7b5d	P03	BOGOF	472	1902
	0efde2	P03	BOGOF	448	1895
	07e4ea	P04	BOGOF	486	1890
	855f49	P04	BOGOF	481	1890
	0c9f61	P03	BOGOF	433	1883
	6d65d2	P04	BOGOF	480	1867
	4a8c38	P04	BOGOF	469	1861
	84a2f4	P04	BOGOF	423	1801
	693708	P03	BOGOF	454	1788
	05e55f	P03	BOGOF	415	1759
	726ac4	P04	BOGOF	454	1756
	368ca0	P04	BOGOF	432	1736
	caa1e1	P03	BOGOF	432	1736
	572031	P03	BOGOF	423	1734
	141d98	P03	BOGOF	387	1695
	284539	P03	BOGOF	384	1678

Question 3:

DISTINCT Promotion Types

Query:

```
SELECT DISTINCT promo_type FROM fact_events;
```

Output:

	promo_type
▶	50% OFF
	25% OFF
	BOGOF
	500 Cashback
	33% OFF

Question 4:

Basic Aggregation

Query:

```
SELECT COUNT(event_id) AS total_events, SUM(`quantity_sold(before_promo)`) AS  
total_quantity_before, SUM(`quantity_sold(after_promo)`) AS total_quantity_after, AVG(base_price) AS  
avg_base_price, MAX(base_price) AS max_base_price, MIN(base_price) AS min_base_price FROM  
fact_events;
```

Output:

	total_events	total_quantity_before	total_quantity_after	avg_base_price	max_base_price	min_base_price
▶	1500	209050	435473	551.9667	3000	50

Question 5:

Sales Volume by Promotion Type

Query:

```
SELECT promo_type, COUNT(event_id) AS total_events, SUM(`quantity_sold(before_promo)`)  
AS total_before, SUM(`quantity_sold(after_promo)`) AS total_after FROM fact_events GROUP  
BY promo_type ORDER BY total_after DESC;
```

Output:

	promo_type	total_events	total_before	total_after
▶	BOGOF	500	58180	215253
	33% OFF	200	63321	90576
	500 Cashback	100	22299	63180
	25% OFF	400	44007	38290
	50% OFF	300	21243	28174

Question 6: Promotion Uplift

Query:

```
SELECT
    promo_type,
    SUM(`quantity_sold(before_promo)`) AS total_before,
    SUM(`quantity_sold(after_promo)`) AS total_after,
    (SUM(`quantity_sold(after_promo)`) - SUM(`quantity_sold(before_promo)`)) AS quantity_change
FROM fact_events
GROUP BY promo_type
ORDER BY quantity_change DESC;
```

Output:

	promo_type	total_before	total_after	quantity_change
▶	BOGOF	58180	215253	157073
	500 Cashback	22299	63180	40881
	33% OFF	63321	90576	27255
	50% OFF	21243	28174	6931
	25% OFF	44007	38290	-5717

Question7:

Product Performance

Query:

```
SELECT
    p.product_code,
    p.product_name,
    p.category,
    SUM(f.`quantity_sold(after_promo)`) AS total_quantity_after
FROM fact_events f
INNER JOIN dim_products p
    ON f.product_code = p.product_code
GROUP BY p.product_code, p.product_name, p.category
ORDER BY total_quantity_after DESC;
```

Output:

	product_code	product_name	category	total_quantity_after
►	P04	Atliq_Farm_Chakki_Atta (1KG)	Grocery & Staples	81290
	P03	Atliq_Suflower_Oil (1L)	Grocery & Staples	74478
	P15	Atliq_Home_Essential_8_Product_Combo	Combo1	63180
	P02	Atliq_Sonamasuri_Rice (10KG)	Grocery & Staples	53235
	P01	Atliq_Masoor_Dal (1KG)	Grocery & Staples	37341
	P13	Atliq_High_Glo_15W_LED_Bulb	Home Appliances	29928
	P14	Atliq_waterproof_Immersion_Rod	Home Appliances	23685
	P07	Atliq_Curtains	Home Care	16317
	P08	Atliq_Double_Bedsheet_set	Home Care	15058
	P12	Atliq_Lime_Cool_Bathing_Bar (125GM)	Personal Care	10280
	P10	Atliq_Cream_Beauty_Bathing_Soap (12...	Personal Care	7697
	P11	Atliq_Doodh_Kesar_Body_Lotion (200ML)	Personal Care	7022
	P09	Atliq_Body_Milk_Nourishing_Lotion (120...	Personal Care	6505
	P05	Atliq_Scrub_Sponge_For_Dishwash	Home Care	4985
	P06	Atliq_Fusion_Container_Set_of_3	Home Care	4472

Question 8:

Category-Level Performance

Query:

```
SELECT
    p.category,
    COUNT(f.event_id) AS event_count,
    SUM(f.`quantity_sold(before_promo)`) AS total_before,
    SUM(f.`quantity_sold(after_promo)`) AS total_after,
    (SUM(f.`quantity_sold(after_promo)`) - SUM(f.`quantity_sold(before_promo)`)) AS quantity_change
FROM fact_events f
JOIN dim_products p
    ON f.product_code = p.product_code
GROUP BY p.category ORDER BY total_after DESC;
```

Output:

	category	event_count	total_before	total_after	quantity_change
►	Grocery & Staples	400	126970	246344	119374
	Combo1	100	22299	63180	40881
	Home Appliances	200	14713	53613	38900
	Home Care	400	19764	40832	21068
	Personal Care	400	25304	31504	6200

Question 9: Store Performance

Query:

```

SELECT
    s.city,
    COUNT(f.event_id) AS event_count,
    SUM(f.`quantity_sold(before_promo)`) AS total_before,
    SUM(f.`quantity_sold(after_promo)`) AS total_after
FROM fact_events f
JOIN dim_stores s
    ON f.store_id = s.store_id
GROUP BY s.city
ORDER BY total_after DESC;

```

Output:

	city	event_count	total_before	total_after
►	Bengaluru	300	49171	105141
	Chennai	240	39505	83273
	Hyderabad	210	34363	69399
	Coimbatore	150	18150	38900
	Mysuru	120	18569	37470
	Visakhapatnam	150	17175	33916
	Madurai	120	14458	31169
	Mangalore	90	7529	14929
	Vijayawada	60	5297	11106
	Trivandrum	60	4833	10170

Question 10:

Campaign Performance

Query:

```
SELECT
    c.campaign_name,
    c.start_date,
    c.end_date,
    COUNT(f.event_id) AS event_count,
    SUM(f.`quantity_sold(before_promo)`) AS total_before,
    SUM(f.`quantity_sold(after_promo)`) AS total_after
FROM fact_events f
JOIN dim_campaigns c
    ON f.campaign_id = c.campaign_id
GROUP BY c.campaign_id, c.campaign_name, c.start_date, c.end_date
ORDER BY total_after DESC;
```

Output:

	campaign_name	start_date	end_date	event_count	total_before	total_after
▶	Sankranti	2024-01-10	2024-01-16	750	98731	252069
	Diwali	2023-11-12	2023-11-18	750	110319	183404

Question 11:

Product Category with HAVING

Query:

```
SELECT
    p.category,
    SUM(f.`quantity_sold(after_promo)`) AS total_quantity_after,
    AVG(f.base_price) AS avg_base_price
FROM fact_events f
JOIN dim_products p
    ON f.product_code = p.product_code
GROUP BY p.category
HAVING SUM(f.`quantity_sold(after_promo)`) > 1000
ORDER BY total_quantity_after DESC;
```

Output:

	category	total_quantity_after	avg_base_price
▶	Grocery & Staples	246344	385.0000
	Combo1	63180	3000.0000
	Home Appliances	53613	685.0000
	Home Care	40832	490.0000
	Personal Care	31504	102.3750

Question 12:

Store + Category Analysis

Query:

```
SELECT
    s.city,
    p.category,
    SUM(f.`quantity_sold(after_promo)`) AS total_quantity_after
FROM fact_events f
JOIN dim_stores s
    ON f.store_id = s.store_id
JOIN dim_products p
    ON f.product_code = p.product_code
GROUP BY s.city, p.category
ORDER BY s.city ASC, total_quantity_after DESC;
```

Output:

	city	category	total_quantity_after
►	Bengaluru	Grocery & Staples	59431
	Bengaluru	Combo1	15250
	Bengaluru	Home Appliances	12909
	Bengaluru	Home Care	10006
	Bengaluru	Personal Care	7545
	Chennai	Grocery & Staples	46639
	Chennai	Combo1	12081
	Chennai	Home Appliances	10686
	Chennai	Home Care	7974
	Chennai	Personal Care	5893
	Coimbatore	Grocery & Staples	22735
	Coimbatore	Combo1	5369
	Coimbatore	Home Appliances	4606
	Coimbatore	Home Care	3527
	Coimbatore	Personal Care	2663
	Hyderabad	Grocery & Staples	39488
	Hyderabad	Combo1	9337
	Hyderabad	Home Appliances	8466
	Hyderabad	Home Care	6691

Question 13:

Promotion Effectiveness by Product

Query:

```
SELECT

    p.product_name,

    p.category,

    SUM(f.`quantity_sold`before_promo) AS total_before,

    SUM(f.`quantity_sold`after_promo) AS total_after,

    (SUM(f.`quantity_sold`after_promo) - SUM(f.`quantity_sold`before_promo)) AS quantity_change,

    ((SUM(f.`quantity_sold`after_promo) - SUM(f.`quantity_sold`before_promo)) /

    NULLIF(SUM(f.`quantity_sold`before_promo), 0)) * 100 AS percentage_change

FROM fact_events f

JOIN dim_products p

    ON f.product_code = p.product_code

GROUP BY p.product_code, p.product_name, p.category

ORDER BY percentage_change DESC;
```

Output:

	product_name	category	total_before	total_after	quantity_change	percentage_change
►	Atliq_waterproof_Immersion_Rod	Home Appliances	6468	23685	17217	266.1874
	Atliq_High_Glo_15W_LED_Bulb	Home Appliances	8245	29928	21683	262.9836
	Atliq_Double_Bedsheet_set	Home Care	4203	15058	10855	258.2679
	Atliq_Curtains	Home Care	4592	16317	11725	255.3354
	Atliq_Home_Essential_8_Product_Combo	Combo1	22299	63180	40881	183.3311
	Atliq_Farm_Chakki_Atta (1KG)	Grocery & Staples	32340	81290	48950	151.3605
	Atliq_Suflower_Oil (1L)	Grocery & Staples	31309	74478	43169	137.8805
	Atliq_Masoor_Dal (1KG)	Grocery & Staples	26040	37341	11301	43.3986
	Atliq_Sonamasuri_Rice (10KG)	Grocery & Staples	37281	53235	15954	42.7939
	Atliq_Doodh_Kesar_Body_Lotion (200ML)	Personal Care	5257	7022	1765	33.5743
	Atliq_Lime_Cool_Bathing_Bar (125GM)	Personal Care	7718	10280	2562	33.1951
	Atliq_Cream_Beauty_Bathing_Soap (125...	Personal Care	6380	7697	1317	20.6426
	Atliq_Body_Milk_Nourishing_Lotion (120ML)	Personal Care	5949	6505	556	9.3461
	Atliq_Scrub_Sponge_For_Dishwash	Home Care	5762	4985	-777	-13.4849
	Atliq_Fusion_Container_Set_of_3	Home Care	5207	4472	-735	-14.1156

Question 14:

Campaign and Promotion Type Analysis

Query:

```
SELECT
    c.campaign_name,
    f.promo_type,
    COUNT(f.event_id) AS event_count,
    SUM(f.`quantity_sold(before_promo)`) AS total_before,
    SUM(f.`quantity_sold(after_promo)`) AS total_after,
    (SUM(f.`quantity_sold(after_promo)`) - SUM(f.`quantity_sold(before_promo)`)) AS quantity_change
FROM fact_events f
JOIN dim_campaigns c
    ON f.campaign_id = c.campaign_id
GROUP BY c.campaign_name, f.promo_type
ORDER BY c.campaign_name ASC, quantity_change DESC;
```

Output:

	campaign_name	promo_type	event_count	total_before	total_after	quantity_change
►	Diwali	500 Cashback	50	16791	50769	33978
	Diwali	BOGOF	200	10024	34461	24437
	Diwali	33% OFF	100	29152	43117	13965
	Diwali	50% OFF	200	16843	22074	5231
	Diwali	25% OFF	200	37509	32983	-4526
	Sankranti	BOGOF	300	48156	180792	132636
	Sankranti	33% OFF	100	34169	47459	13290
	Sankranti	500 Cashback	50	5508	12411	6903
	Sankranti	50% OFF	100	4400	6100	1700
	Sankranti	25% OFF	200	6498	5307	-1191

Question 15:

Product Revenue Before and After Promotion

Query:

```
SELECT
    p.product_name,
    p.category,
    SUM(f.base_price * f.`quantity_sold(before_promo)`) AS revenue_before,
    SUM(f.base_price * f.`quantity_sold(after_promo)`) AS revenue_after,
    (SUM(f.base_price * f.`quantity_sold(after_promo)`) - SUM(f.base_price *
f.`quantity_sold(before_promo)`) AS revenue_difference
FROM fact_events f
JOIN dim_products p
    ON f.product_code = p.product_code
GROUP BY p.product_code, p.product_name, p.category
ORDER BY revenue_difference DESC;
```

Output:

	product_name	category	revenue_before	revenue_after	revenue_difference
►	Atliq_Home_Essential_8_Product_Combo	Combo1	66897000	189540000	122643000
	Atliq_Farm_Chakki_Atta (1KG)	Grocery & Staples	10851800	29100500	18248700
	Atliq_waterproof_Immersion_Rod	Home Appliances	6597360	24158700	17561340
	Atliq_Sonamasuri_Rice (10KG)	Grocery & Staples	32061660	45782100	13720440
	Atliq_Double_Bedsheet_set	Home Care	5001570	17919020	12917450
	Atliq_Sunflower_Oil (1L)	Grocery & Staples	5599512	14310708	8711196
	Atliq_High_Glo_15W_LED_Bulb	Home Appliances	2885750	10474800	7589050
	Atliq_Curtains	Home Care	1377600	4895100	3517500
	Atliq_Masoor_Dal (1KG)	Grocery & Staples	4478880	6422652	1943772
	Atliq_Doodh_Kesar_Body_Lotion (200ML)	Personal Care	998830	1334180	335350
	Atliq_Lime_Cool_Bathing_Bar (125GM)	Personal Care	478516	637360	158844
	Atliq_Cream_Beauty_Bathing_Soap (125...	Personal Care	393625	483145	89520
	Atliq_Body_Milk_Nourishing_Lotion (120ML)	Personal Care	601270	671830	70560
	Atliq_Scrub_Sponge_For_Dishwash	Home Care	316910	274175	-42735
	Atliq_Fusion_Container_Set_of_3	Home Care	2160905	1855880	-305025

Question 16:

Classify Promotion Performance

Query:

```
SELECT
    promo_type,
    SUM(`quantity_sold(before_promo)`) AS total_before,
    SUM(`quantity_sold(after_promo)`) AS total_after,
    ((SUM(`quantity_sold(after_promo)`) - SUM(`quantity_sold(before_promo)`)) /
    NULLIF(SUM(`quantity_sold(before_promo)`), 0)) * 100 AS percentage_change,
    CASE
        WHEN ((SUM(`quantity_sold(after_promo)`) - SUM(`quantity_sold(before_promo)`)) /
        NULLIF(SUM(`quantity_sold(before_promo)`), 0)) * 100 >= 50 THEN 'High Impact'
        WHEN ((SUM(`quantity_sold(after_promo)`) - SUM(`quantity_sold(before_promo)`)) /
        NULLIF(SUM(`quantity_sold(before_promo)`), 0)) * 100 >= 20 THEN 'Medium Impact'
        ELSE 'Low Impact'
    END AS performance_category
FROM fact_events
GROUP BY promo_type
ORDER BY percentage_change DESC;
```

Output:

	promo_type	total_before	total_after	percentage_change	performance_category
►	BOGOF	58180	215253	269.9777	High Impact
	500 Cashback	22299	63180	183.3311	High Impact
	33% OFF	63321	90576	43.0426	Medium Impact
	50% OFF	21243	28174	32.6272	Medium Impact
	25% OFF	44007	38290	-12.9911	Low Impact

Question 17:

Top Products Within Each Category

Query:

WITH product_rankings AS (

SELECT

p.category,

p.product_name,

SUM(f.`quantity\_sold(after\_promo)`) AS total_quantity_after,

DENSE_RANK() OVER (

PARTITION BY p.category

ORDER BY SUM(f.`quantity\_sold(after\_promo)`) DESC

) AS category_rank

FROM fact_events f

JOIN dim_products p

ON f.product_code = p.product_code

GROUP BY p.category, p.product_code, p.product_name

)

SELECT

category,

product_name,

total_quantity_after,

category_rank

FROM product_rankings

WHERE category_rank <= 2

ORDER BY category, category_rank;

Output:

	category	product_name	total_quantity_after	category_rank
	Combo1	Atliq_Home_Essential_8_Product_Combo	63180	1
	Grocery & Staples	Atliq_Farm_Chakki_Atta (1KG)	81290	1
	Grocery & Staples	Atliq_Sunflower_Oil (1L)	74478	2
	Home Appliances	Atliq_High_Glo_15W_LED_Bulb	29928	1
	Home Appliances	Atliq_waterproof_Immersion_Rod	23685	2
	Home Care	Atliq_Curtains	16317	1
	Home Care	Atliq_Double_Bedsheet_set	15058	2
	Personal Care	Atliq_Lime_Cool_Bathing_Bar (125GM)	10280	1
	Personal Care	Atliq_Cream_Beauty_Bathing_Soap (125GM)	7697	2

Question 18:

Best-Performing Stores Within Each City

Query: (Ai help used)

WITH store_rankings AS (

SELECT

s.city,

f.store_id,

SUM(f.`quantity\_sold(after\_promo)`) AS total_quantity_after,

DENSE_RANK() OVER (

PARTITION BY s.city

ORDER BY SUM(f.`quantity\_sold(after\_promo)`) DESC

) AS city_rank

FROM fact_events f

JOIN dim_stores s

ON f.store_id = s.store_id

GROUP BY s.city, f.store_id

)

SELECT

city,

store_id,

total_quantity_after,

city_rank

```
FROM store_rankings
WHERE city_rank <= 2
ORDER BY city, city_rank;
```

Output:

	city	store_id	total_quantity_after	city_rank
▶	Bengaluru	STBLR-7	11866	1
	Bengaluru	STBLR-6	11601	2
	Chennai	STCHE-4	11547	1
	Chennai	STCHE-7	11546	2
	Coimbatore	STCBE-0	8657	1
	Coimbatore	STCBE-2	8535	2
	Hyderabad	STHYD-2	10822	1
	Hyderabad	STHYD-0	10522	2
	Madurai	STMDU-0	8310	1
	Madurai	STMDU-2	7685	2
	Mangalore	STMLR-2	5253	1
	Mangalore	STMLR-1	5187	2
	Mysuru	STMYS-1	11773	1
	Mysuru	STMYS-3	9833	2
	Trivandrum	STTRV-0	5193	1
	Trivandrum	STTRV-1	4977	2
	Vijayawada	STVJD-0	5751	1
	Vijayawada	STVJD-1	5355	2
	Visakhanat...	STVSK-1	7614	1

Question 19:

Campaign-Level Product Performance

Query: (used AI Help with this)

```
WITH campaign_product_stats AS (
SELECT
    c.campaign_name,
    p.product_name,
    SUM(f.`quantity_sold(before_promo)`) AS total_before,
    SUM(f.`quantity_sold(after_promo)`) AS total_after,
    (SUM(f.`quantity_sold(after_promo)`) - SUM(f.`quantity_sold(before_promo)`)) AS
quantity_change,
    ((SUM(f.`quantity_sold(after_promo)`) - SUM(f.`quantity_sold(before_promo)`)) /
    NULLIF(SUM(f.`quantity_sold(before_promo)`), 0)) * 100 AS percentage_change,
```

```

DENSE_RANK() OVER (
    PARTITION BY c.campaign_name
        ORDER BY ((SUM(f.`quantity_sold(after_promo)`) - SUM(f.`quantity_sold(before_promo)`)) /
NULLIF(SUM(f.`quantity_sold(before_promo)`), 0)) * 100 DESC
    ) AS campaign_rank
FROM fact_events f
JOIN dim_campaigns c
    ON f.campaign_id = c.campaign_id
JOIN dim_products p
    ON f.product_code = p.product_code
GROUP BY c.campaign_id, c.campaign_name, p.product_code, p.product_name
)
SELECT
    campaign_name,
    product_name,
    total_before,
    total_after,
    quantity_change,
    percentage_change,
    campaign_rank
FROM campaign_product_stats
WHERE campaign_rank <= 3
ORDER BY campaign_name, campaign_rank;

```

Output:

	campaign_name	product_name	total_before	total_after	quantity_change	percentage_change	campaign_rank
►	Diwali	Atliq_waterproof_Immersion_Rod	2015	6950	4935	244.9132	1
	Diwali	Atliq_Curtains	2680	9214	6534	243.8060	2
	Diwali	Atliq_High_Glo_15W_LED_Bulb	3215	11053	7838	243.7947	3
	Sankranti	Atliq_Suflower_Oil (1L)	16257	61185	44928	276.3610	1
	Sankranti	Atliq_waterproof_Immersion_Rod	4453	16735	12282	275.8141	2
	Sankranti	Atliq_High_Glo_15W_LED_Bulb	5030	18875	13845	275.2485	3

Question 20:

Complete Promotional Performance Analysis

Query: (used AI Help with this)

```
WITH product_performance AS (  
    SELECT  
        p.product_name,  
        p.category,  
        COUNT(f.event_id) AS event_count,  
        SUM(f.`quantity_sold(before_promo)`) AS total_before,  
        SUM(f.`quantity_sold(after_promo)`) AS total_after,  
        (SUM(f.`quantity_sold(after_promo)`) - SUM(f.`quantity_sold(before_promo)`)) AS  
quantity_change,  
        ((SUM(f.`quantity_sold(after_promo)`) - SUM(f.`quantity_sold(before_promo)`)) /  
NULLIF(SUM(f.`quantity_sold(before_promo)`), 0)) * 100 AS percentage_change,  
        SUM(f.base_price * f.`quantity_sold(before_promo)`) AS revenue_before,  
        SUM(f.base_price * f.`quantity_sold(after_promo)`) AS revenue_after,  
        (SUM(f.base_price * f.`quantity_sold(after_promo)`) - SUM(f.base_price *  
f.`quantity_sold(before_promo)`)) AS revenue_change,  
        AVG(f.base_price) AS avg_base_price,  
        DENSE_RANK() OVER (  
            PARTITION BY p.category  
                ORDER BY (SUM(f.base_price * f.`quantity_sold(after_promo)`) - SUM(f.base_price *  
f.`quantity_sold(before_promo)`)) DESC  
        ) AS product_rank  
    FROM fact_events f  
    JOIN dim_products p  
        ON f.product_code = p.product_code  
    GROUP BY p.category, p.product_code, p.product_name  
)
```

```
SELECT product_name, category, event_count, total_before, total_after, quantity_change,
percentage_change, revenue_before, revenue_after, revenue_change, avg_base_price, product_rank
FROM product_performance WHERE product_rank <= 2 ORDER BY category, product_rank;
```

Output:

product_name	category	event_count	total_before	total_after	quantity_change	percentage_change	revenue_before	revenue_after	revenue_change	avg_base_price	product_rank
Atliq_Home_Essential_8_Product_Combo	Combo 1	100	22299	63180	40881	183.3311	66897000	189540000	122643000	3000.0000	1
Atliq_Farm_Chakkis_Atta (1KG)	Grocery & Staples	100	32340	81290	48950	151.3605	10851800	29100500	18248700	330.0000	1
Atliq_Sonamasuri_Rice (10KG)	Grocery & Staples	100	37281	53235	15954	42.7939	32061660	45782100	13720440	860.0000	2
Atliq_waterproof_Immersion_Rod	Home Appliances	100	6468	23685	17217	266.1874	6597360	24158700	17561340	1020.0000	1
Atliq_High_Glo_15W_LED_Bulb	Home Appliances	100	8245	29928	21683	262.9836	2885750	10474800	7589050	350.0000	2
Atliq_Double_Bedsheet_set	Home Care	100	4203	15058	10855	258.2679	5001570	17919020	12917450	1190.0000	1
Atliq_Curtains	Home Care	100	4592	16317	11725	255.3354	1377600	4895100	3517500	300.0000	2
Atliq_Doodh_Kesar_Body_Lotion (200ML)	Personal Care	100	5257	7022	1765	33.5743	998830	1334180	335350	190.0000	1
Atliq_Lime_Cool_Bathing_Bar (125GM)	Personal Care	100	7718	10280	2562	33.1951	478516	637360	158844	62.0000	2